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Palaeomagnetism of Permian multiple intrusion dykes in Bohuslän, SW Sweden

Leif Thorning* and Niels Abrahamsen *Laboratory of Geophysics, University of Aarhus, Finlandsgade 8, DK-8200 Aarhus N, Denmark*

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Summary. A detailed rock and palaeomagnetic investigation of the three major dyke systems of late Palaeozoic age in Bohuslän SW Sweden is reported. The three dyke systems are dolerite dykes, porphyritic dolerite dykes and rhomb porphyry (RP) dykes, the latter ones being of special interest as examples of multiple intrusion dykes. The rhomb porphyry dykes are closely related to the rhomb porphyry lavas and dykes of the Permian igneous province of the Oslo Rift zone some 100–200 km NNW. Geometrical mean values of natural remanence and susceptibility are of the order of 10^{-3} emu cm^{-3}, and the Q-ratio is usually < 1. Magnetic fabric indicates vertical flow of the magma, and discrepancies in the Curie-temperatures when measured and calculated by microprobe analysis, respectively, suggest the very stable primary component of the TCRM to reside in single domain titanomagnetites exsolved in larger grains where a soft VRM dominates. Detailed sampled profiles across two well developed multiple intrusion RP dykes show a marked consistency in the symmetrical build-up of the dykes and the magnetic parameters (bulk susceptibility, Q-ratio, anisotropy and coercivity spectrum), which strongly supports the theory of the multiple intrusion mechanism of these composite RP dykes.

During af cleaning, the remanent direction of the RP dykes migrates along great circles and stabilizes around $(D, I) = (199°, -32°)$ $\alpha_{95} = 6.6°$, $N = 8$ sites, with the apparent pole position 46°S, 345°E, which is very close to other poles of lower Permian age from stable Europe. In the dolerites and porphyritic dolerites VRMs of very low coercivities dominate, and a stable direction was difficult to isolate by af demagnetization. Because of the higher scatter the cleaned mean directions of the dolerite dykes and porphyritic dolerite dykes however are not significantly different from that of the rhomb porphyry dykes, and all dyke types are considered to be of approximately the same age.

* Now at: The Geological Survey of Greenland, Øster Voldgade 10, DK-1350 Copenhagen K, Denmark.

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1 Introduction

Dykes equivalent to rock types of the Permian Oslo region (S Norway) are known to occur far outside the graben zone of the rift system.

The purpose of the present paper is to summarize the results of palaeomagnetic and rock-magnetic investigations of such dykes in Bohuslän, SW Sweden (Fig. 1). They are believed to be of Permian age too, although no radiometric datings have yet been published.

At least two generations of dykes older than the Permian ones are known in the Bohuslän area. To the NW the numerous and conspicuous dykes of the Koster islands are found being older than the 910 Myr old Bohus Granite (Skiöld 1976). To the south in the vicinity of Gothenburg, the palaeomagnetic direction of WNW-striking dykes are equivalent to ages around 800–900 Myr (Abrahamsen 1974). Finally, a younger group of minor ultrabasic dykes is found, which may be of lower Palaeozoic age (Samuelsson, personal communication).

2 Geology

Dykes belonging to the lower Permian igneous rock complex of the Oslo region (Fig. 1(a)) are known more than 100 km away from the downfaulted graben zone to NW, SW and SE (Oftedal 1952, 1960). A great number of such dykes are further known from the Swedish Skagerrak coast (Fig. 1(b)), most of them being vertical and with a strike between N–S and NNW–SSE, i.e. within the main trend of the Oslo Rift (N–S) and the Tornquist line of the Fennoscandian border zone (NW–SE), TL (Ramberg *et al.* 1977). The bedrock of Bohuslän is built up of entirely Precambrian Bohus Granite and gneisses. The degree of exposure, however, is low, Quaternary deposits making the connection between the dyke localities dubious except in a few cases of major dykes.

Among several investigations of the dykes the most detailed description has been given by Ljungner (1927). Recently, Samuelsson (1967a, 1971, 1973) made a detailed investigation on the relationship between the narrow dolerites (D), the larger porphyritic dolerites (PD), and the rhomb prophyry dykes (RP), the latter ones being multiple intrusives with marginal dolerites (Dm), marginal rhomb porphyry (RPm) and central rhomb porphyry (RPc). Because of their intriguing multiple intrusive mode, the RP-dykes are of special interest, wherefore they were further made subject to a more detailed rock magnetic investigation (Thorning 1973).

The description of geology and petrology below is based primarily on Samuelsson (1971).

2.1 DOLERITES (D) AND PORPHYRITIC DOLERITES (PD)

These dykes (Fig. 1) may be porphyritic or non-porphyritic. Generally, the latter are narrow, whereas the porphyritic dolerites may attain a width of ~10 m. One of them has been traced some 30 km (site 14 and 17, Fig. 1).

The dolerites are dark, fine to medium grained basaltic rocks with plagioclase, pyroxene, and secondary epidote, sericite, chlorite, leucoxene, and calcite; the opaques (≤ 10 per cent) are mainly titanomagnetites and some pyrite.

The mineralogy of the PD's resembles that of the dolerites apart from the plagioclase phenocrysts, the sizes of which are typically around 0.5 cm. The margin is usually fine-grained and non-porphyritic, and the density of the phenocrysts generally increases towards the centre.

The non-porphyritic margins of the PD's have chilled contacts to the wall rock, whereas no chilled contact is found between the non-porphyritic margins and the central porphyritic part, but the latter contact zone may contain xenoliths. This may indicate a two-step intrusion mode of the PD's: First a non-porphyritic narrow marginal dolerite with xenoliths

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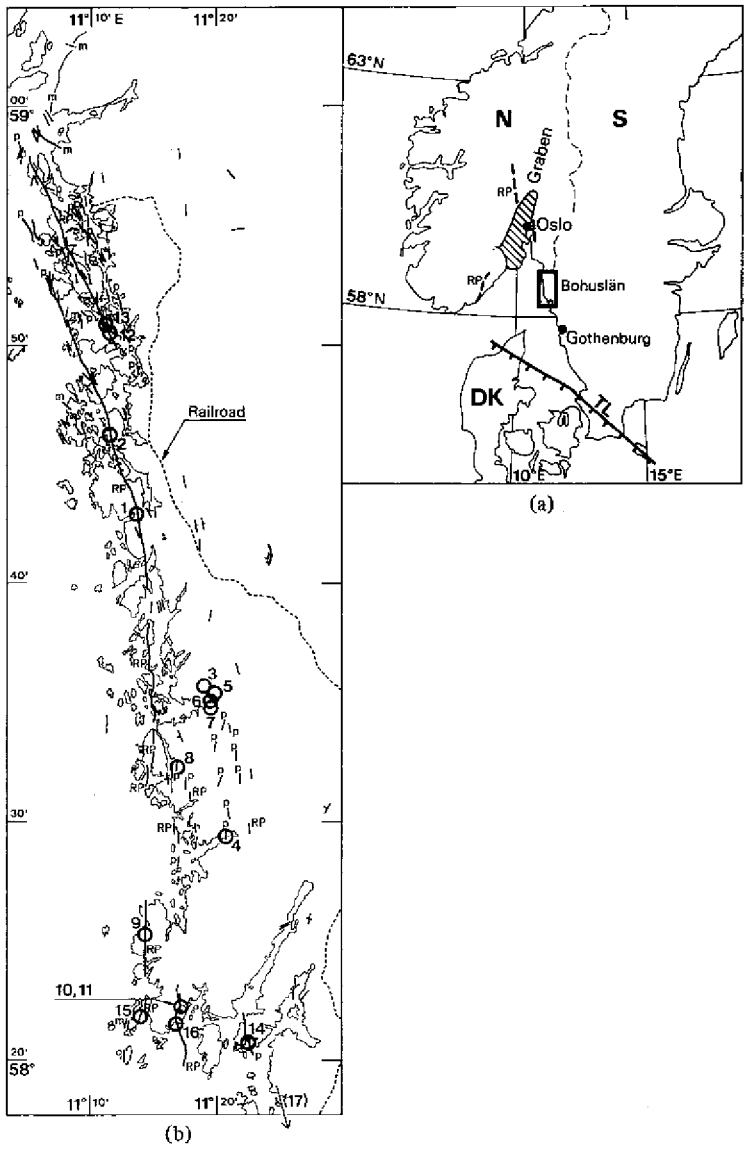


Figure 1. (a) Top: index map showing the area investigated (Bohuslän), S.E. of the Permian igneous province of the Oslo Graben. RP indicates rhomb porphyric dykes outside the graben area. TL is the Tornquist line separating the stable Scandinavian Shield from the Phanerozoic sedimentary basins to the S.W. (b) Left: coastal area of Bohuslän with the palaeomagnetic sampling sites 1–16; site 17 (Lyrön) is 30 km south of site 14. RP (including RPc, RPm and Dm) indicates rhomb porphyry dykes, p indicates porphyritic dolerites PD, and m ultramafic dykes, while the dolerites D are without letter (redrawn from Asklund 1947). The bedrock is entirely Precambrian.

in a central zone was intruded. Before this zone had been wholly crystallized, the dyke opened again and the magma of the porphyritic dolerite was intruded.

2.2 RHOMB PORPHYRY DYKES (RP)

These dykes occur over quite a large area along the Skagerrak coast (Fig. 1). The longest one known is some 30 km, and the width of the dykes vary from 5 to 50 m. They exhibit

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interesting petrological variations and resemble the rhomb porphyry lavas and dykes of the Oslo field.

A grain size difference between the marginal and central parts of these dykes already noted by Ljunger (1927), and Samuelsson (1967a, 1971) describes the types in detail. The structural relationships are shown schematically in Fig. 2. Each of the three rock types is very homogeneous over the entire region.

The colour of the Dm is dark-grey to black. The contact to the surrounding Precambrian granite is always chilled, and granite xenoliths showing signs of assimilation may often be found.

The RPm consists of rhomb-shaped crystals of feldspars in a matrix dominated by elongated crystals of zoned plagioclase, but also containing a variety of other minerals (pyroxene, amphibole, biotite, apatite, chlorite, serpentine, titanomagnetite and pyrite). The matrix of the RPc is more fine-grained than that of the RPm, and consists of plagioclase, alkalifeldspar, pyroxene, chlorite, serpentine, quartz, amphiboles, apatite, titanomagnetite and pyrite. The rock is grey in colour (red when weathered), with numerous rhomb-shaped feldspars of greyish-white colour. The contact between the two types of rhomb porphyry is usually marked by a 2–3 cm thin zone of lighter colour with hardly any phenocrysts. The RPc may be slightly chilled towards the RPm.

2.3 GENESIS OF THE DYKES

Based on structural and petrological observations, Samuelsson (1967a, 1971) suggested the following hypotheses for the origin and emplacement of the dykes: In a cooling acid magma, where the crystallization had not yet been completed, rhomb-shaped phenocrysts of feldspars and pyroxenes existed in aggregates. Another basaltic magma was also present, probably situated in a reservoir beneath the acid magma, and at a certain time it intruded into the acid magma, with the result of mixing the two magmas; it also intruded the surrounding granite, thereby forming the dolerite dykes.

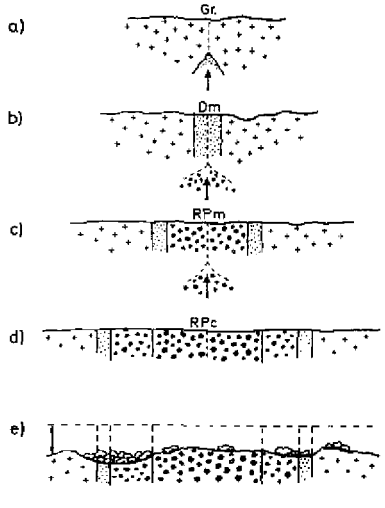


Figure 2. Stages in the stepwise evolution of the composite multiple intrusive RP dykes. After intrusion and cooling of the marginal dolerite Dm, the marginal rhomb porphyry RPm was intruded and cooled, and finally the central rhomb porphyry RPc was intruded.

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Into some of the dolerite dykes two other intrusion impulses followed soon, resulting in the marginal and central rhomb porphyries, respectively, the latter types representing different stages of magma mixing or different levels in the magma reservoir.

The mode of formation of the porphyritic dolerites is not quite clear, but we may suggest that they were formed by an analogue mixing of the basaltic magma with a feldspar cumulus from the acid magma.

While the relative age of the composite RP-dykes thus appears to be Dm–RPm–RPc (youngest), the relative ages between D, PD and RP are not clear, as no cuttings have been observed. Accepting the above hypothesis of two co-existing magmas in Permian times, the ages of the individually intruded dykes, however, are not likely to be very different. In a discussion of the Oslo region, Oftedahl (1952, 1960) suggested a likely co-existence of different magma types and the possibility of mixing.

With the exception of a possible block-faulting related to the Fennoscandian border zone and glacial isostatic movements, the area does not appear to have experienced any significant tectonic disturbances since the time of intrusion of the dykes.

2.4 INTRUSION LEVEL AND COOLING TIME

Considering the general lack of Phanerozoic deposits in the area, the casual occurrence of a quartzitic boulder (Asklund 1947) in one of the RP-dykes (site 10, Vägga), and the numerous clastic 'dykes' of Palaeozoic fissure fillings in the Precambrian basement surface now exposed (Samuelsson 1975), the initial depth of the present erosion level and hence the intrusion level of the Permian dykes is likely to be of the order of 1 km and certainly less than say 3 km. This constrains the initial temperature of the surrounding bedrock at the time of intrusion to less than 100°C even with allowance for a moderately elevated regional geothermal gradient in the Permian period. A rough estimate of the cooling times required for the temperature of the dykes to pass below the relevant blocking temperatures for the carriers of the remanent magnetization may then be made by solving the equation of thermal diffusion (Jaeger 1968). Taking roughly the initial temperatures of the magma and the bedrock to be 1100 and 100°C, respectively, and the thermal diffusivity of both to be around 0.01 $cm^2 s^{-1}$, a dyke 10 m wide would cool below 500, 400, 300 and 200°C within about 1.2, 2.5, 6 and 25 yr, respectively. For other dykes the cooling times are proportional to the squares of the thicknesses relative to the 10 m thick dyke; thus a 3.33 m dyke cools nine times faster, a 30 m dyke cools nine times slower, all other parameters being equal.

As most of the dykes are about 10 m wide, the thicker ones all being multiple intrusive RP dykes, we conclude that the stable remanent direction of each dyke, rather than an average

Table 1. Rock magnetic parameters J, κ, Q.

Type	n	J^* (10^{-3} emu cm^{-3})	Range*	n	κ^* (10^{-3} G Oe^{-1})	Range*	n	$\bar{Q}$	Range
Dm	6	1.61	(0.9–2.5)	6	7.39	(6.2–8.5)	6	0.46	(0.26–0.71)
RPm	3	1.64	(1.4–1.9)	4	4.65	(3.7–5.2)	3	0.67	(0.54–0.84)
RPc	8	0.62	(0.14–1.4)	7	2.59	(1.7–3.4)	7	0.63	(0.37–0.97)
RP	17	1.03	(0.14–2.5)	17	4.30	(1.7–8.5)	16	0.57	(0.26–0.97)
PD	3	2.85	(0.9–12.6)	1	3.37	(2.7–4.3)†	1	7.48	
D	6	2.17	(1.5–6.2)	5	5.45	(4.4–6.2)	5	0.66	(0.56–0.95)

* Between site logarithmic means.

† Within site logarithmic means (site 4, Gerlesborg).

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direction, is likely to represent a short-time spot reading of the geomagnetic field with very little smoothing of a possible secular variation.

In case of the multiple intrusive dykes, the elder (marginal) parts must have been reheated and were at least partially remagnetized in accordance with the remanence of the younger central part of the dyke. Thus, in the two cases (sites 2 and 16, Table 1), where the thickness of the central and marginal parts are known, the reheating of the margin would amount to an increase in temperature of at least about 200 and 300°C, respectively. This reheating would tend to smooth out any minor differences in the directions of the remanence.

3 Magnetic investigations

3.1 SAMPLING

Samples were mostly collected by means of a portable drill and in some cases as hand samples (sites 4, 14, 15 and 17), and the orientations were performed with a sun compass. About 200 individually oriented cores or samples were collected from 17 sites, eight of them being from rhomb porphyry dykes. At site 2 (Raftötungan) the cores were taken in a detailed profile (II) across the entire width of the dyke with samples for every 50–70 cm. At some of the other RP-sites, less complete profiles were also taken (Table 2).

3.2 TECHNIQUES APPLIED

The NRM and anisotropy of susceptibility (of RP) was measured with a Digico® Spinner Magnetometer (Molyneux 1971), and the stability of the remanence was investigated in a detailed programme of af demagnetization in zero field (±10γ). Thermal demagnetization was attempted, but not successful. The bulk susceptibility was measured *in situ* and in the laboratory using bridge type instruments (Collinson & Molyneux 1967). Susceptibility versus temperature was also studied on powder samples (Stephenson & de Sa 1970).

Polished sections of RP-specimens were examined under the ore microscope and with a Geoscan microprobe. The magnetic minerals were measured for their Fe and Ti contents and rough counts were performed for Mn, Al, Mg and Cr.

3.3 INTENSITY OF NRM, SUSCEPTIBILITY AND Q-VALUES

In Table 1 the logarithmic mean value and range of the bulk susceptibility κ and the intensity J of the NRM are summarized together with the Q-values ($Q = J/\kappa H$) of the rock types investigated. The NRM mean intensity of the D's and PD's are about twice that of the RP-dykes and more scattered. The bulk susceptibility of the RP-dykes decreases from the margin (Dm) towards the centre (RPc), and that of PD is lower than D, the latter being slightly higher than that of the RP. The mean of the Q-values ranges between 1/4 and 1 except at site 4 (Gerlesborg), the PD of which was curiously high ($\bar{Q}=7$).

The mean magnetic properties thus generally show decreasing values of κ and J with increasing acidity, which was to be expected.

3.4 ANISOTROPY OF SUSCEPTIBILITY

A detailed study of the susceptiblity and its anisotropy was made on the three rock types of the well-developed composite RP-dyke of site 2 (Raftötungan). The considerable variation across the vertical dyke is shown in detail in Fig. 3. The degree of anisotropy ($An = \kappa_{max}/\kappa_{min}$

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Table 2. NRM and af cleaned palaeomagnetic results

Site no.	Locality, profile	Grid	Type	Width (m)	NRM					af cleaned						
					N	D (°)	I (°)	α_{95} (°)	k	N	mdf (Oe)	$\bar{F}$ (Oe)	D (°)	I (°)	α_{95} (°)	k	R
1	Manneklåvan, I	295104	RPc + RPm (+ Dm)	30	11	197	61	11	17	8		400	195	−25	9.8	32	7.7849
2	Raftötungan, II	270168	RPc + RPm (+ Dm)	6 + 3 × 2 + 0.6 × 2	24	185	71	9	11	34	50–200	225	200	−32	4.4	32	32.9730
3	Edsten 1	347975	D (?PD)	11	10	94	87	14	14	4	50	100–200	168	−32	24	16	3.8089
4	Gerlesborg	372859	PD	11	7	195	−45	56	2.1	8	30–200	100–300	199	−53	24	6.5	6.9193
5	Edsten 2	349969	D	10	8	257	72	15	15	8	< 50	150–300	252	−46	23	6.7	6.9599
6	Edsten 3	348968	D (?PD)	6	8	95	83	32	3.9	6	≪ 50	300	188	−82	46	3.0	4.3835
7	Edsten 4	348967	D	11	7	183	84	19	11	5	< 50	150	163	−38	79	1.6	3.0158
8	Slottet, Hambgsd.	330909	D (?PD)	⩾ 6	9	280	50	31	3.6	3	50–150	200–300	230	−46	21	35	2.9430
9	Varpeberget, IV	309782	RPc + Dm	~ 50	12	229	75	6	62	11		300	199	−28	7.4	38	10.7402
10	Vägga Nord, V + VI	337719	RPc + RPm (+ Dm)	> 45	23	217	81	9	11	12		400	200	−31	7.3	36	11.6908
11	Vägga Syd	336717	RPc (+ Dm)		8	214	54	11	27	8		300	207	−40	8.1	47	7.8526
12	Långö Syd, VII	266252	RP + Dm	5 + 0.5	9	189	11	36	3.0	12		225	194	−49	4.1	110	11.8997
13	Långö Nord	264256	RP (+ Dm)		6	226	25	74	1.7	4		500	194	−25	7.3	158	3.9810
14	Näverkärr	390701	PD	⩾ 6	8	258	42	60	1.8	8	50–200	50–200	194	−50	12	21	7.6652
15	Hasselösund	309717	D	1	9	181	−6	39	2.6	8	50–200	100	192	−37	8.9	40	7.8254
16	Vägga Kyst, Smögen	337712	RPc + RPm + Dm	19 + 5 × 2 + 0.7 × 2	12	185	71	5	76	12		300	204	−24	7.1	38	11.7126
17	Lyrön	497403	PD	~ 10	10	177	70	9	32	10	50–150	100–400	182	−31	8.8	31	9.7125

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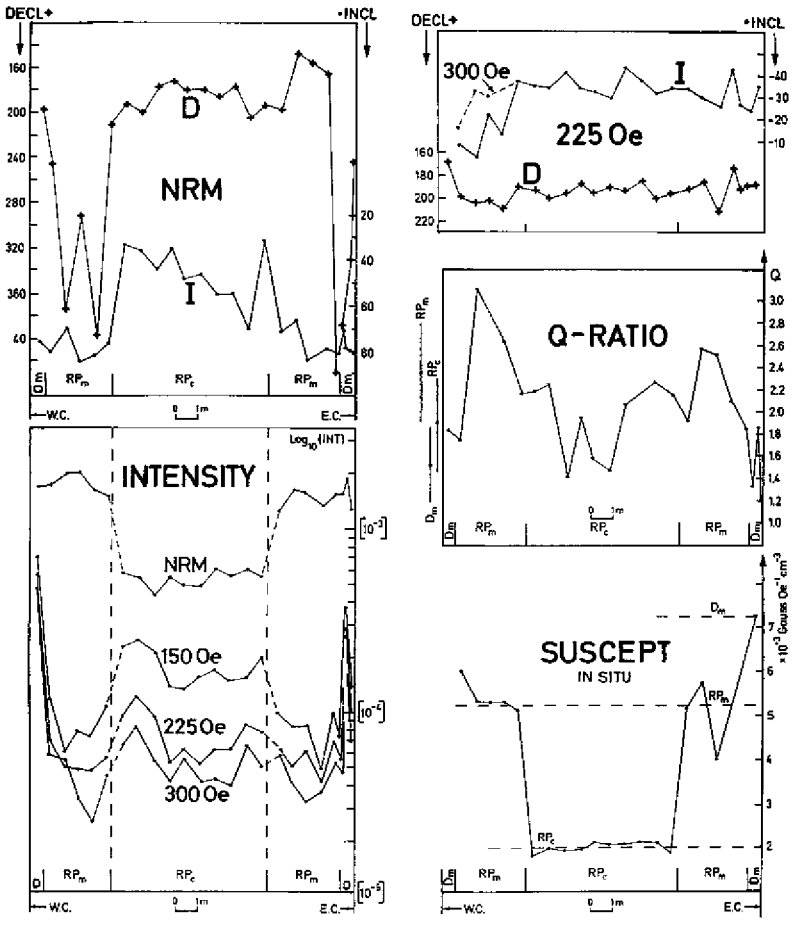


Figure 3. Systematic and symmetrical variation of the magnetic parameters in profile II across the well developed composite rhomb porphyry dyke at Raftötungan. The three susceptibility levels clearly distinguish between the three rock types involved: Dm, RPm and RPc, as does the differential response of the remanent intensities to af demagnetization.

is also varying across the dyke (Fig. 4), but there are no marked differences between the three rock types. The directions of the axes of the anisotropy ellipsoid are also plotted in Fig. 4. It is striking that the axis of maximum susceptibility (a) is well defined and nearly exactly vertical for all three rock types. The intermediate and minimum axes (b and c) are not well defined for the marginal dolerite, while the b and c axis of the two varieties of rhomb porphyry are close to each other.

3.5 MICROPROBE ANALYSIS AND CURIE TEMPERATURES

From profile II at Raftötungan 12 polished sections were prepared, and 138 analyses on 98 grains were examined in the microprobe.

The diameter of the grains examined varied from 20–500μm, the lower limit being set by the electron beam (spot readings with a 10–15μm diameter beam). Smaller grains were

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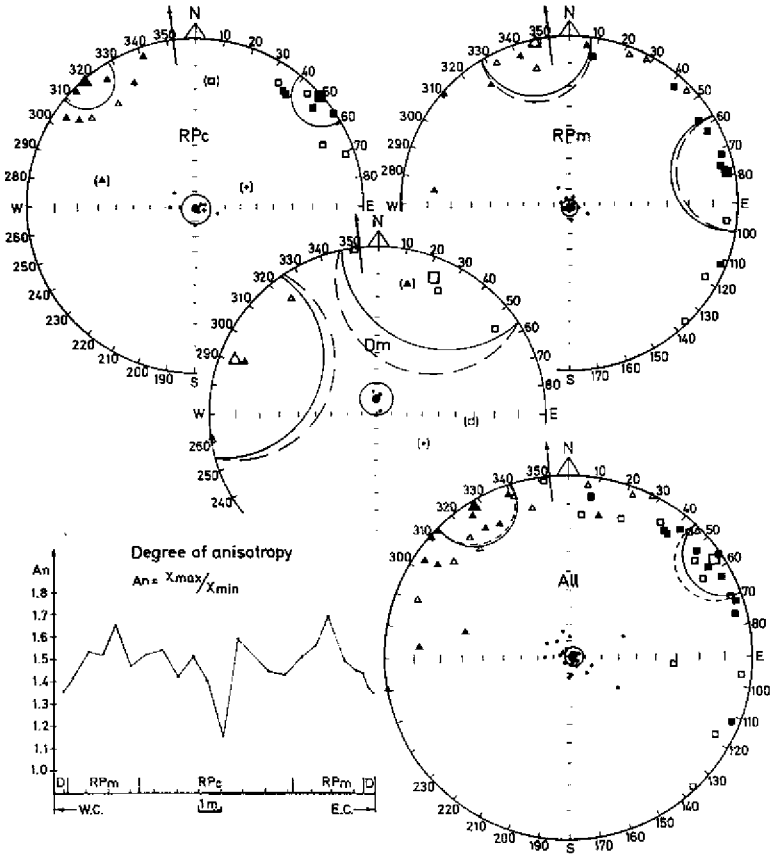


Figure 4. Profile II (Raftötungan): degree of anisotropy and orientation of anisotropy ellipsoid (stereographic projection) for the individual rock types of the composite dyke. The arrow indicates the strike of the dyke. Dots, triangles and squares are directions of maximum (a), intermediate (b) and minimum (c) susceptibility axes, respectively. The predominantly vertical flow of the magma is suggested by the vertical elongation of the major magnetic grains, as determined by the tight clustering of the a-axis of the magnetic fabric. The strike of the dyke does not appear to have influenced the orientation of the intermediate and minimum axis, illustrating that the magnetic grains have equidimensional rather than tabular cross sections orthogonal to the vertical (longer) axis.

numerous too. No correlation between grain size, grain form, composition and rock type could, however, be observed on this scale, the overall picture being that of a very complex assemblage of magnetic grains. The compositions of a number of titanomagnetite and haemoilmenite phases were calculated, and on the assumption of a linear relationship between composition and Curie temperatures T_c the corresponding T_c were calculated. In Fig. 5 these are given by numbers and plotted together with the observed T_c.

Thermomagnetic curves (Fig. 6) were further measured on 21 powder samples in air from profile II. Most of the samples exhibited only one T_c of 550–570°C, but a few samples also had a break on the thermo-magnetic curve in the interval between 350–400°C, indicating the presence of an additional phase with that T_c (arrows in Fig. 5). No clear difference between the three rock types can be seen, neither are calculated T_c of the three rock types significantly different. There is, however, a distinct discrepancy between the calculated and the observed T_c as discussed later.

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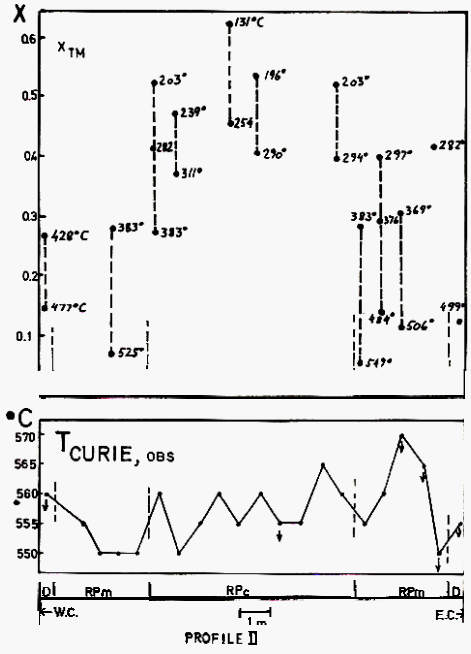


Figure 5. Microprobe analysis of profile II of the Ti-content in titanomagnetites expressed by the variable x, according to the formula $x\,Fe_2TiO_4 \cdot (1-x)Fe_3O_4$. Further is shown the observed T_C as measured on powdered specimens and determined from plots of reversible low-field susceptibility versus temperature curves.

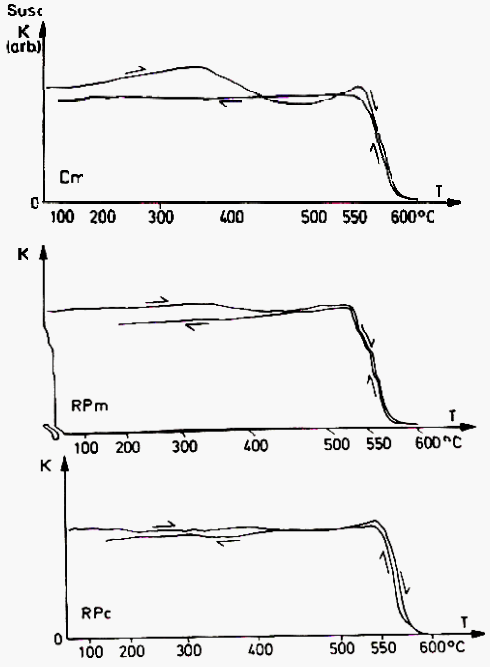


Figure 6. Typical examples of susceptibility versus temperature (in air) of powder specimens of Dm, RPm and RPc.

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3.6 REMANENT MAGNETIZATION OF THE RHOMB PORPHYRY DYKES

The site mean direction of the NRM's are shown in Fig. 7(b). They seem to be smeared along a vertical great circle with a declination of 190–210°. At site 2 this is caused mainly by a significant difference of inclination of the RPm and Dm on one side and the RPc on the other. This was found to be a general feature although the total means of each rock type is not significantly different. The direction and intensity of the NRM at Raftötungan are also plotted in detail in Fig. 3.

The Q-ratio varies across the dykes, and is in profile II (Fig. 3) generally more than 1.

Four pilot specimens of each RP type were initially af demagnetized in eleven to seventeen steps up to maximum fields of 1100 to 1700 Oe peak (Fig. 8a, b and c). The Dm-specimens did not give any other indication of a stable remanent magnetization than the NRM, the RPm yielded a stable component of magnetization with negative inclination, and

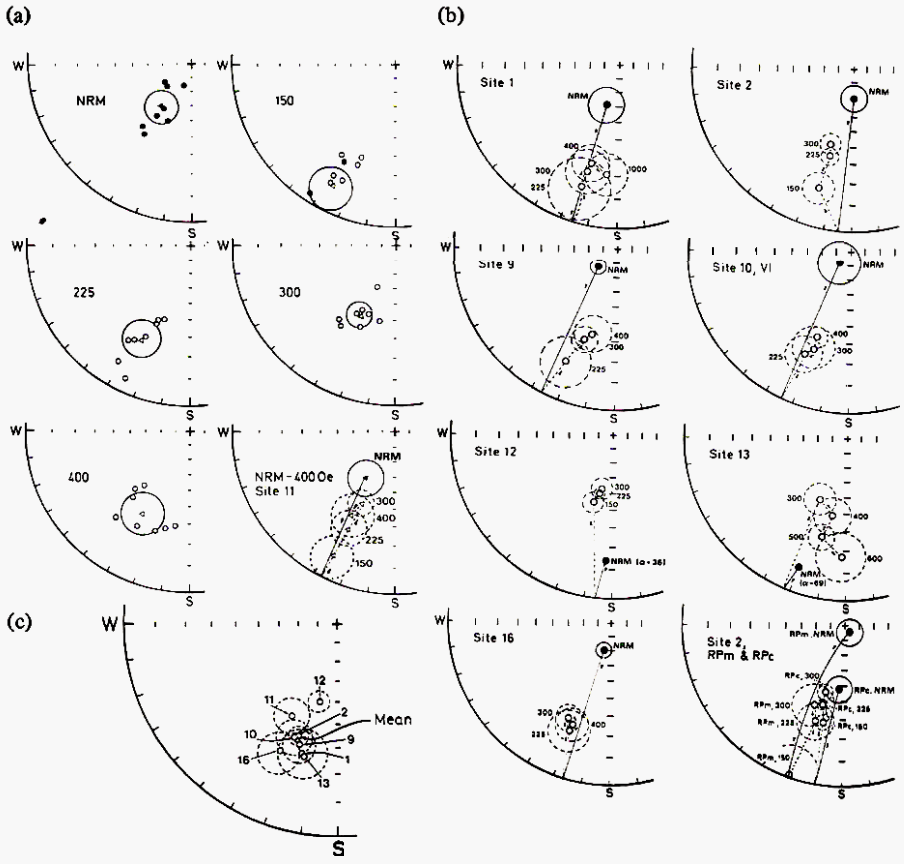
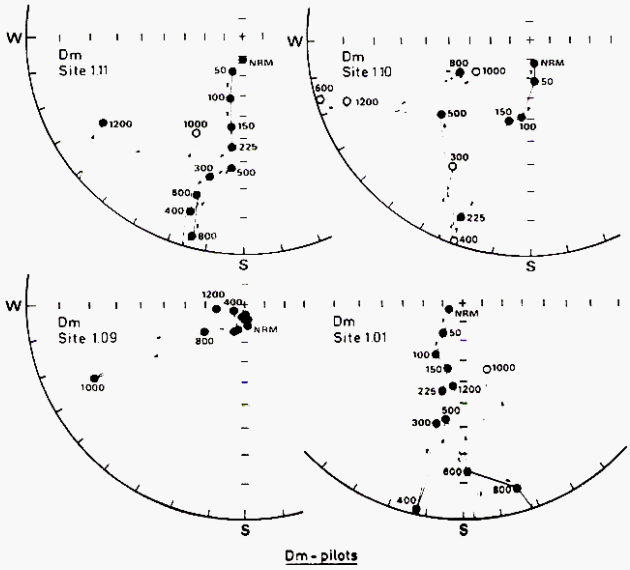


Figure 7. Stereographic plots of the remanent magnetization of the RP sites after stepwise af demagnetization in zero field. Black dots lower hemisphere (positive inclination), open dots upper hemisphere (negative inclination). Numbers indicate peak af magnetic fields. (a) Site 11. Directions of stepwise demagnetized remanent magnetizations of individual specimens, with α_{95} circles (cones of confidence) shown. The last figure shows the former five combined. (b) Change of site mean directions of all individual RP-sites (except site 11) during af demagnetization. Site 2 (Raftötungan) is shown combined as well as split up in marginal (RPm) and central (RPc) rhomb porphyry specimens, respectively. (c) af cleaned site mean directions of all RP-sites with 95 per cent significance circles indicated. All cleaned RP mean directions show negative inclinations between −24 and −49° and SSW declinations, and are interpreted as the primary TRM or TCRM of the dykes.

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the RPc showed a similar result although certain differences between the RPm and the RPc emerged, which will be touched upon later. The subsequent bulk demagnetization of all specimens in three to five steps up to 400 to 500 Oe confirmed these results (Figs 8 and 9). It was found that the RPm and RPc carry a stable remanent component of the same direction, and its coercivity must be over 1000 Oe. The secondary magnetization is of a much lower coercivity, but makes up a large percentage of the NRM. This is especially so in the case of the RPm. As demonstrated in Fig. 3 the isolated stable remanence of the RPc clearly shows higher coercivities, at least in the performed steps of demagnetization, 150, 225 and 300 Oe peak. Already at 150 Oe the curve is 'inverted', the normalized intensity of the RPc now being the higher. This and the two peaks of intensity near the contact between RPm and RPc will be discussed later. Fig. 7(c) shows the mean direction of the cleaned remanence, accepted as the best estimate of the primary magnetization. Only in a few cases it was possible to isolate a stable direction of remanence in the marginal dolerite of approximately the same direction as in the rhomb porphyries. Table 2 gives the mean values and Fisher statistics (Fisher 1953) of the demagnetization results.

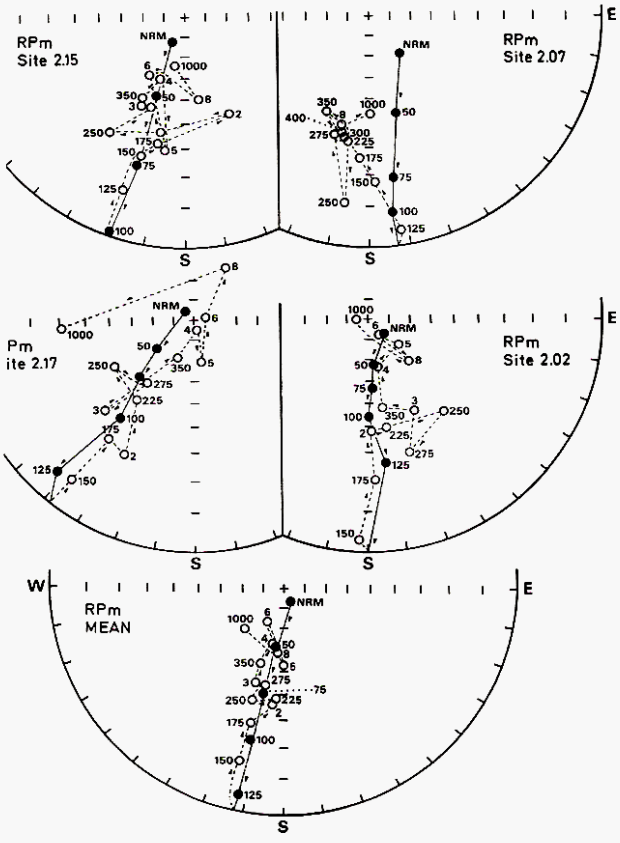
The af cleaned RP-dykes thus yield a well defined and stable characteristic direction of high coercivity remanent magnetization, which within the experimental errors are the same for the three rock types. However, the scatter of the Dm-directions was considerably higher than that of the RPc and RPm, and it was difficult to isolate a stable remanence of the Dm.



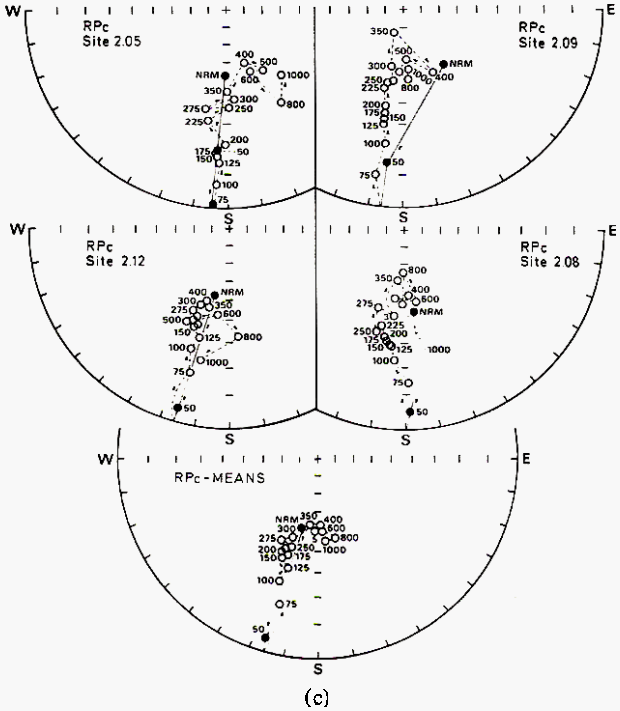
(a)

Figure 8. Detailed af demagnetization experiments with individual Dm, RPm and RPc specimens. Stereographic projection of remanent magnetization as in Fig. 7. (a) Four Dm pilot specimens. Small numbers indicate peak af demagnetizing field. No stable end points are defined, but the great circle paths are generally equivalent at low fields to those of the RP-specimens. Above 300 to 500 Oe spurious components are dominating the directions. (b) Four RPm pilot specimens. The great circle paths during af demagnetizations are well defined, and fairly stable directions are found between 200 and 500 Oe. (c) RPc pilot specimens. The great circle paths are well defined, and stable directions are found above *c*. 200 Oe.

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(b)



(c)

Figure 8 – *continued*

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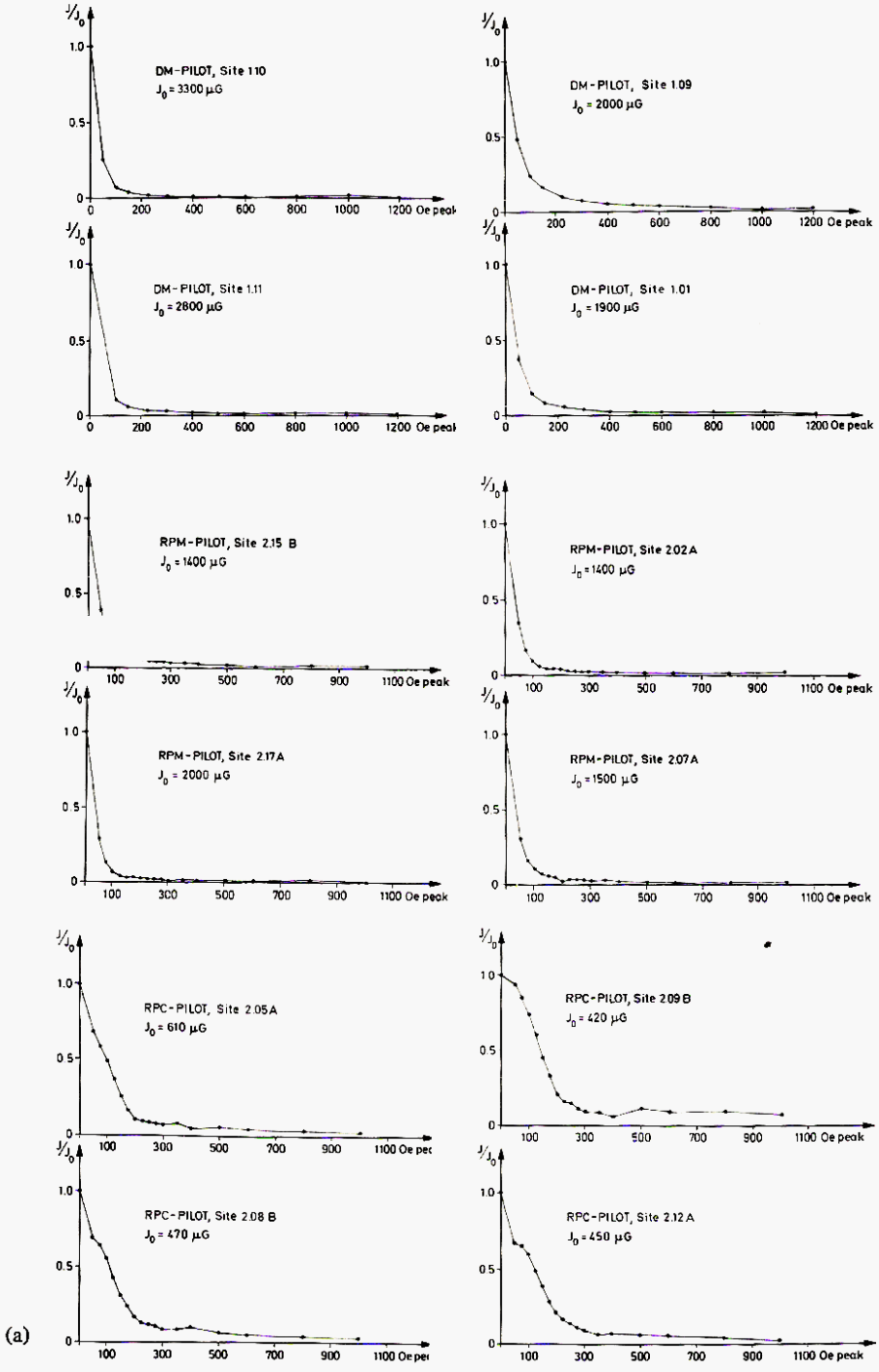


Figure 9. (a) Decay of the remanent intensity during af demagnetization of the same pilot specimens as in Fig. 8(a), (b) and (c). The decay curves of the RPc pilots are significantly different from those of the Dm and RPm specimens. (b) Decay curves of the remanent magnetization normalized against the intensity of 150, 125 and 100 Oe for the Dm, RPm and RPc, respectively. High coercivity components are present in all cases, but in the Dm the direction of the stable (supposed primary) component is drowned in the spurious components above 400 Oe, *cf.* Fig. 8(a).

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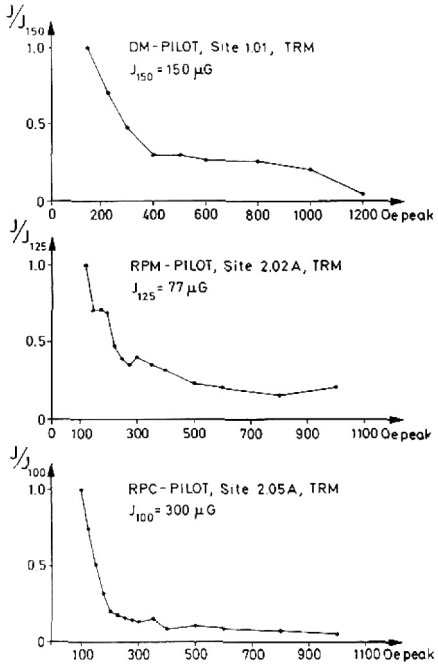


Figure 9 (b)

3.7 REMANENT MAGNETIZATION OF THE DOLERITE AND PORPHYRITIC DOLERITE DYKES

After pilot af demagnetizations of one specimen from each site in steps of 100–200 Oe up to 1400 Oe, all specimens were stepwise demagnetized up to 500 Oe peak, and in some cases this range was extended when a characteristic direction was difficult to define. In all such cases only spurious magnetizations were encountered with. In Table 2 the NRM directions and the af cleaned (edited) results together with Fisher statistics are summarized.

In most cases the mdf (median destructive field) was very low (typically $\lesssim$ 50 Oe), a viscous component with steeply inclined positive inclination dominating in all sites but site 4 (which had a high Q and intensity of the NRM, the volume percentage of opaque minerals being 7 per cent, Samuelsson 1971). This explains the difficulties involved in isolating a characteristic stable direction and also why the scatter is generally high, and not usually improved by the cleaning process.

An attempt was made to distinguish between the dolerite and porphyritic dolerite dykes, but no significant differences of the RMs were found, although the means of the characteristic directions are somewhat better defined in the porphyritic dolerites (k = 11.4 in Table 3) than in the dolerites (k = 5.0). This may be a qualitative rock magnetic argument that the porphyritic dolerites occupy an intermediate compositional position between the dolerites and the rhomb porphyry rocks (with overall precision parameters of k = 18.5 and 36.5 for the RPm and RPc, respectively) as previously suggested.

4 Rock magnetic properties in relation to the multiple intrusion theory

The RP dykes exhibit a high degree of magnetic anisotropy, the direction of easiest magnetization is vertical with small scatter, whereas the intermediate and minimum susceptibility axes are less well defined. This is probably due to a preferred vertical orientation of

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the elongated magnetic grains (and other grains) resulting from the predominantly vertical flow of the magma at the time of intrusion. The anisotropy is probably mainly of the type associated with the larger grains present in the rocks. The higher and equal scatter of the b and c axes may indicate that the grains are of ellipsoidal rather than tabular shape, and subsequent observations in the ore microscope supported this. Thus the pressure-determined interactions with the contacts of the intrusion cannot have been of great effect, and the magma flow must have been the dominant factor. These results confirm the utility of magnetic anisotropy as a petrofabric indicator.

To understand the nature and composition of the remanent magnetization of the dykes the following should be borne in mind, as demonstrated for the RP dykes:

(1) Two main components of magnetization in the RP dykes have been found – a stable component of high coercivity with a negative inclination and far from the direction of any recent geomagnetic field, and an unstable component in the direction of easiest magnetization with a positive inclination and close to the present geomagnetic field.

(2) The observed anisotropy probably originates from the accumulated effects of the shape anisotropy of the larger elongated magnetic grains in the rocks. A large number of smaller grains are also present.

(3) The observed T_c indicate that the dominant magnetic phase has a T_c of 550–560°C. This is a titanomagnetite with only a little Ti substituted. There are also indications of another titanomagnetite phase with a T_c of 350–400°C.

(4) The microprobe proved that besides titanomagnetite almost pure ilmenite is also present.

(5) There is a discrepancy between the observed and the calculated T_c, the latter usually being 100–200°C lower.

The overall picture is more complicated, but in the main the following conclusions seem reasonable:

The NRM of the RP dykes consists of a stable primary component and an unstable secondary component. The secondary component has been induced by recent geomagnetic fields and has been deflected into the direction of easiest magnetization. This component is carried by the larger, multi-domain magnetic grains, and so it is of low coercivity. The very hard primary magnetization consequently must be associated with small single-domain or pseudo-single-domain grains of almost pure magnetite, which are too small to be examined in the microprobe. The primary magnetization may also be carried by lamellae of Fe-rich titanomagnetites exsolved in the larger grains. This is supported by the discrepancy demonstrated between calculated and observed T_c. Such exsolution process takes place on a scale much too fine for the microprobe to distinguish between the resulting individual lamellae of magnetite and ilmenite. The readings would then be averages of the two phases, and the calculated T_c would be too low. The magnetic phase with a T_c of 300–400° C observed from the thermomagnetic curves could be the mother phase of the exsolution process (Creer & Ibbertson 1970).

It is worth recalling the very complex conditions under which the primary magnetization originated. Disregarding for a moment the possibility of repeated heatings due to several intrusion impulses, the highly complicated environment and thermochemical processes of a magma cooling in a dyke should still be taken into consideration. The magnetization is therefore likely to be carried by a number of grains of different origin; however, the primary remanence is of the thermoremanent or thermochemical remanent type, and it originated during the relatively short period of time while the dykes were cooling below the respective blocking temperatures. Demagnetization effects due to the shape of a dyke may result in

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significant rotation of the induced field into the plane of the dyke (Strangway 1961). However, the declination of the geomagnetic field in Permian times was quite close to the strike of the dykes, and the possible maximum error due to this effect would be in the range 1–2° of the declination. Furthermore, because the anisotropy is mainly related to the preferred flow pattern of the multi-domain grains, whereas an alignment of the small single-domain grains is less likely to occur, it is concluded that the primary magnetization is hardly influenced by the anisotropy. Therefore the isolated primary component probably is a good and true representation of the local geomagnetic field at the cooling time. The palaeomagnetic dating performed in the following discussion is consequently justified. Finally, a baked contact test on a few samples from the much older granite very near the dykes supports this conclusion. The direction of the NRM in the granite, albeit weak, was almost identical to that of the isolated primary magnetization of the RP dykes, indicating a parallel remagnetization of the isotropic granite at the time of the dyke intrusions.

From Samuelsson (1971) it is known that the total amount of iron is highest in the Dm and lowest in the RPc. The variation of susceptibility shown in Fig. 3 is in obvious agreement with this and furthermore, places the boundary within 1–2 cm of the contact between the rock types. It is difficult to imagine how these sharp boundaries could have been formed by some differentiation process (an alternative explanation of the formation of the dykes). The mechanism of multiple intrusion seems to be the much more likely explanation.

Further support of this theory can be found in the way the intensity profile changes during af demagnetization (Fig. 3). The violent change of intensity again outlines the contact between the two rhomb porphyries. The primary magnetization makes up a much larger percentage of the RPc than of the RPm, suggesting a different composition and/or history of the two RP types.

In the presentation of the results the attention was drawn to the two small peaks of intensity in the RPc very near the RPc/RPm boundary. The higher intensity at this exact location can be explained by a larger number of small magnetic grains, i.e. the rock has been chilled. The same effect to be expected in the RPm near the contact to the Dm is hidden, since the intensity of the Dm is always higher than that of the RPm.

Other differences were revealed during af cleaning. Whilst the primary magnetization of the RPc was easily recovered, the RPm offered more resistance, and the Dm only seldom showed any signs of a primary magnetization. These differences may be due to the number of re-heatings the different rock types have experienced, as these would have direct influence on the acquisition of the primary magnetization. The re-heatings could also influence the rock types indirectly through the maybe reduced resistance of the re-heated rocks to the later effects of, for instance, weathering.

5 Palaeomagnetic poles and ages

In Table 3 mean directions, Fisher statistics (Fisher 1953) and resulting palaeopoles of the af cleaned stable remanent magnetization of the different dyke types are compiled, together with the combined mean directions D + PD and RP + PD + D, all the stable components being reversely magnetized.

The mean directions of the respective dyke types are not significantly different (Fig. 10). However, the between site precision parameter of the RP dykes ($k = 72$) is considerably higher than those of the porphyritic dolerites and dolerite dykes, the values of α_{95} being 6.6, 21 and 29°, respectively. The site mean direction of all dykes is Dm = 197° and Im = −38° ($\alpha_{95} = 8.6°, k = 19, N = 16$).

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Table 3. Mean palaeomagnetic results and pole positions.

Rock type	N	D (°)	I (°)	α_{95} (°)	k	R	Pole positions Lat. ° S	Long. ° E	δp (°)	δm (°)
RPc + RPm + Dm	8*	199	−31.8	6.6	71.7	7.9024	46.4	344.5	4.2	7.4
RPc + RPm + Dm	101	198	−33.0	2.6	29.5	97.6211	47.3	345.5	1.7	2.9
RPc	44	198	−32.7	3.6	36.5	42.8244	47.2	345.6	2.3	4.1
RPm	16	196	−27.9	8.8	18.5	15.1935	44.7	349.2	5.2	9.5
Dm										
PD	3*	190	−44.9	20.8	36.2	2.9447	57.1	354.6	16.6	26.3
PD	26	189	−43.5	8.8	11.4	23.8136	56.2	356.5	6.8	11.9
D (excl. site 6)†	5*	198	−45.0	29.1	7.9	4.4929	55.6	341.9	23.2	36.7
D (excl. site 6)	28	199	−44.0	13.6	5.0	22.5768	54.6	340.8	10.7	17.0
D + PD (excl. site 6)	8*	195	−45.1	16.4	12.4	7.4337	56.4	346.5	13.1	20.8
All RP + PD + D (excl. site 6)	16*	197.2	−38.2	8.6	19.4	15.2260	50.9	345.4	6.0	12.7

* Unit weight to site; otherwise unit weight to specimen.
† Site 6 unstable.

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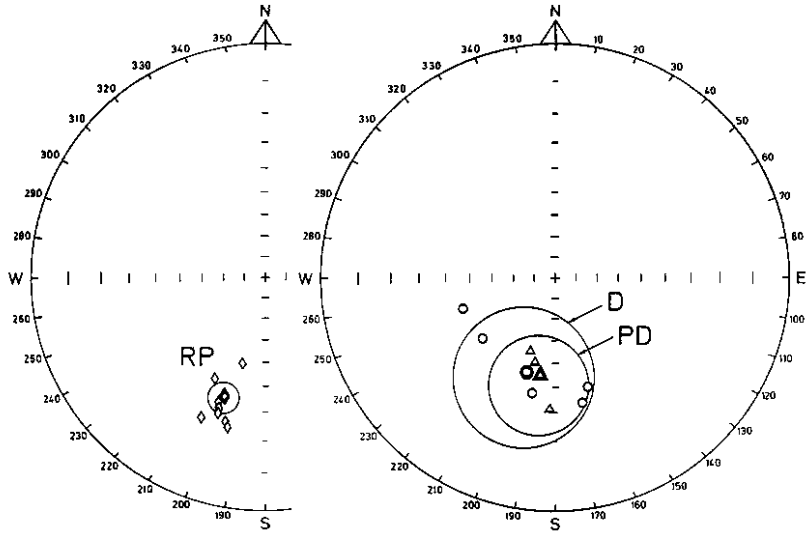


Figure 10. Cleaned mean directions of RP, PD and D, with 95 per cent significance circles. Unit weight to site (stereographic projection).

In Table 4 selected relevant palaeomagnetic poles from Sweden and Norway of palaeozoic ages have been listed. In Fig. 11 these poles are plotted on the present geographical grid together with the site mean (south-seeking) poles of the Bohuslän dykes. The mean pole of all the dyke types (pole 15) and the RP pole (pole 12) are plotted with their site mean 95 per cent significance ovals (full-line ovals), whereas poles of the porphyritic dolerites (pole 13) and dolerites (pole 14) are plotted with broken-line 95 per cent ovals on sample mean levels. The south-seeking mean pole of the RP dykes (46.4° N, 164.5° E, $\delta p = 4.2°$, $\delta m = 7.4°$) is situated about 2° N of the centre of the well clustering Permian poles, the difference being insignificant.

When comparing the RP-pole with the time moving average poles for northern Eurasia (Irving 1977), which are also plotted in Fig. 11 with the 95 per cent-significance circles for every 10 Myr between 300 and 190 Myr, the 95 per cent oval of the site mean RP-pole encloses the mean poles of age 260, 250 and 240 Myr (time scale according to Harland, Smith & Wilcock 1964), suggesting a palaeomagnetic age of the RP-dykes around 250 ± 10 Myr. The well defined mean pole of the Oslo igneous rock complex (pole 10, $\alpha_{95} = 3°$) may be slightly younger, as it coincides with the 240 Myr moving average pole of Irving (1977). Statistically, however, the RP-pole is not significantly different from the Oslo pole.

The primary magnetic direction of the RP-dykes is thus beyond doubt of Permian, and presumably of Lower Permian age (240–280 Myr). This confirms the previous general geological age-correlation of the Bohuslän dykes with the Lower Permian igneous activity of the Oslo Graben.

Poles 4 and 5 (Table 4 and Fig. 11) of the radiometrically dated dolerite sills of Mount Billingen and Hunneberg some 60 km ESE of the present area deviate significantly from the Bohuslän dykes, the K–Ar datings (Mulder 1971) indicating Upper Carboniferous ages. Three poles from other basaltic rocks investigated in S. Sweden (poles 6 and 7) and South Norway (pole 8) fall close to these older poles, whereas pole 9 (melaphyre dykes of Skania) is close to that of the dolerites and porphyritic dolerites.

Finally, it may be mentioned that the deviating pole of the Permian Brummundal lavas in south Norway (pole 11) does not apply in this context, as it is considered to be

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Table 4. Selected palaeomagnetic poles from Sweden and Norway.

Pole	Age (Myr)	Rock unit	N (n)	α_{95} (°)	P lat. ° N	P long. ° E	Reference*
1	Su	Ringerike Sandstone, Norway (60° N, 10° E)	10 (21)		21	159	1, WE 4.1
2	D1	Röragen Sandstone, Norway (62.5° N, 11.9° E)	8 (13)	15	19	160	1, WE 5.1
3	Dm	Kvamshesten Old Red Standstone, Norway (61.4° N, 5.6° E)	(41)	9	22	170	1, WE 5.6
4	Cu (287)	Billingen Sill, Sweden (58.5° N, 14° E)	9 (80)	2	31	174	1, WE 6.15
5	Cu (– P1) (279)	Hunneberg Sill, Sweden (58.5° N, 12.5° E)	3 (33)	5	38	177	1, WE 6.16
6	Cu (P1) (294)	Skaane dolerites, Sweden (55.5° N, 13.5° E)	8 (43)	11	37	174	1, W 6.17, Klingspor (1976), Mulder (1971)
7	Cu	qz-dolerites, Skaane, Sweden (55.5° N, 13.5° E)	13 (106)	6	38	168	Bylund (1974)
8	Cu (P1)	Ny-Hellesund diabase, Norway (58° N, 7.8° E)	10 (37)	3	39	161	1. WE 6.18
9	P1	Melaphyre dykes, Skaane, Sweden (55° N, 13° E)	9 (78)	11	54	172	Bylund (1974)
10	P1 (270)	Oslo igneous complex, Norway (59.7° N, 10.4° E)	27 (494)	3	47	157	1. WE 7.1. van Everdingen (1960)
11	P1 (Mesoz. rem.)	Brummundal lavas, Norway (61° N, 11° E)	3 (12)		48	138	2, 14.319
12	(P1)	Rhomb porphyry dykes, SW Sweden (58.55° N, 11.25° E)	8 (101)	6.6	46.4	164.5	This work
13	(P1)	Porphyritic dolerites, SW Sweden (58.55° N, 11.25° E)	3 (26)	21	57.1	174.6	This work
14	(P1)	Dolerites, SW Sweden (58.55° N, 11.25° E)	5 (28)	29	55.6	161.9	This work
15	(P1)	RP + PD + D, SW Sweden (58.55° N, 11.25° N)	16 (155)	8.6	50.9	165.4	This work

* 1, McElhinny (1973), appendix). 2, McElhinny & Cowley (1977).

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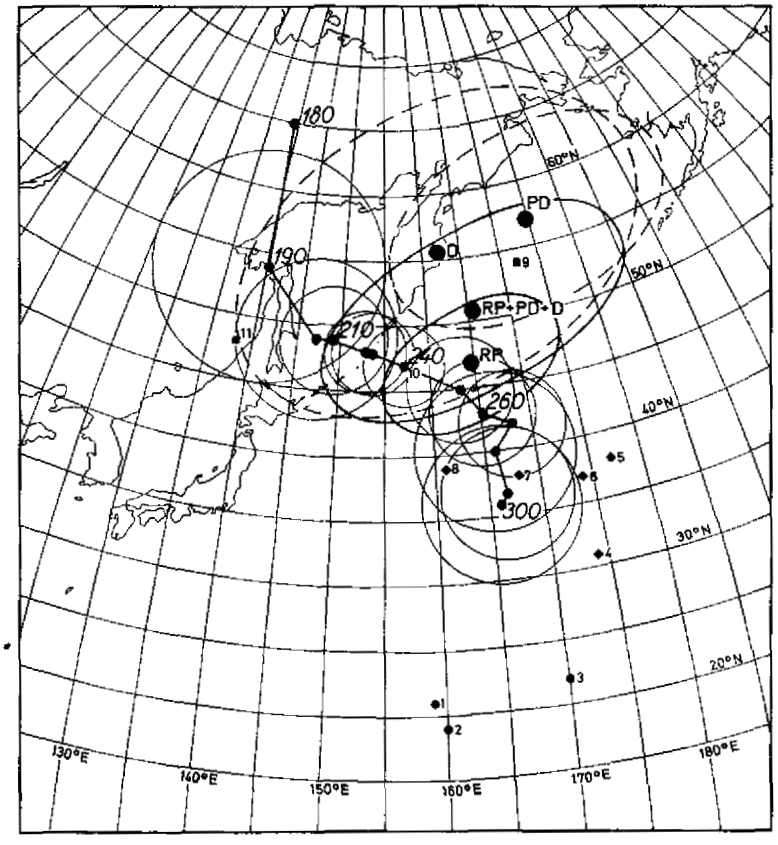


Figure 11. Apparent palaeomagnetic pole positions of selected Palaeozoic rocks from Sweden and Norway, as listed in Table 4, and the 10 Myr time moving average poles for northern Eurasia with 95 per cent significance circles and ages from 300 to 180 Myr (Irving 1977). Lambert equal area projection. Further discussion in text.

remagnetized in Mesozoic time (Storetvedt & Petersen 1970; Storetvedt *et al.* 1978). The reversed mean poles of the Bohuslän dolerites (56° N, 162° E, $\delta p = 23°$, $\delta m = 37°$) and porphyritic dolerites (57° N, 175° E, $\delta p = 17°$, $\delta m = 26°$) both fall about 10° N of the RP-pole and outside the main cluster of Permian poles, although close to pole 9 of the Skania melaphyre dykes. However, due to the higher values of α_{95}, none of these mean pole positions are significantly different from that of the RP-dykes, but differ from the poles 4–8 mentioned above.

We therefore conclude that the dolerite dykes as well as the porphyritic dolerite dykes in Bohuslän are likely to be younger than poles 4–8, and of the same age as the RP-dykes. Since the scatter of the mean poles of the dolerites and porphyritic dolerites is rather high, a possible relative age sequence of the three dyke types cannot be distinguished on the present palaeomagnetic data, as expected from the intrusion theory.

6 Conclusion

Based on the palaeomagnetic and rock magnetic evidence presented above, we conclude:

(1) The magnetic age of the rhomb porphyry dykes in Bohuslän is Lower Permian, and probably equivalent to that of the effusive and hypabyssal basalts and rhomb porphyries

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of the Oslo Graben. The dolerites and porphyritic dolerites in Bohuslän investigated likely to be of the same age, although the primary magnetic direction is less well define these rock types.

(2) The multiple intrusion theory for the formation of the composite rhomb porphy dykes is strongly supported by the rock magnetic properties investigated.

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